

Bridging Stochasticity and Determinism: A Unified Survey of Flow Matching, Trajectory Rectification, and Controllable Generation

Generative modeling is undergoing a paradigmatic shift from stochastic diffusion processes to deterministic flow-based frameworks, unified by the learning of time-dependent vector fields governed by continuity equations [43]. This synthesis examines the convergence of Flow Matching (FM) and Diffusion Models, highlighting how Optimal Transport (OT) theory enables trajectory rectification for efficient, few-step sampling while exposing critical "low-noise pathologies" that destabilize learning near the data manifold [186]. We analyze the tension between geometric straightness and distributional expressivity, contrasting iterative reflow methods with single-stage regularization techniques like Isokinetic Flow Matching [208]. Furthermore, the review addresses the failure of naive guidance in rectified flows, advocating for geometry-aware mechanisms such as Rectified-CFG++ that preserve probability conservation [88]. Finally, we explore the integration of Reinforcement Learning (RL) and optimal control for reward alignment, discussing how frameworks like Flow-GRPO and Flow Map Reward Guidance navigate the trade-off between deterministic efficiency and stochastic exploration [151]. By connecting theoretical guarantees with architectural innovations, this work outlines a roadmap for robust, real-time generative systems capable of complex reasoning and precise control.

Theoretical Convergence of Diffusion and Flow Paradigms

The modern landscape of generative modeling is increasingly defined by a unified mathematical framework that bridges the stochastic nature of diffusion processes with the deterministic mechanics of flow-based transport. At the core of this convergence lies the realization that both diffusion models and flow matching can be understood as instances of learning a time-dependent vector field that governs the evolution of probability densities via continuity or Fokker-Planck equations [43]. Traditionally, score-based generative models rely on Stochastic Differential Equations (SDEs) to define a forward noising process and a corresponding reverse-time SDE for generation, where the reverse drift is determined by the score function of the perturbed data distribution [115]. However, it has been rigorously established that these stochastic dynamics possess an equivalent deterministic counterpart known as the Probability Flow Ordinary Differential Equation (PF-ODE). This ODE yields identical marginal distributions to the SDE at every time step, thereby connecting the stochastic denoising trajectories of diffusion models to the deterministic transport mechanisms characteristic of normalizing flows [115].

Flow Matching (FM) generalizes this paradigm by decoupling the generative path from the specific constraints of diffusion processes. While traditional diffusion models are restricted to Gaussian probability paths defined by variance-preserving or variance-exploding schedules, FM allows for the construction of arbitrary conditional probability paths between a source noise distribution and the target data distribution [13]. In this framework, the model learns a vector field through direct regression on fixed conditional paths, effectively bypassing the need for simulating complex stochastic dynamics during training. This simulation-free approach not only subsumes existing diffusion paths as specific instances but also enables the utilization of more efficient trajectories, such as those derived from Optimal Transport (OT) displacement interpolation, which result in straighter paths and faster sampling [13]. The theoretical underpinning of these methods rests on the continuity equation, which dictates that any valid generative trajectory must satisfy mass conservation as the density evolves over time [19]. Recent work has further refined this by showing that aligning the flow divergence alongside the vector field itself can bound the total variation gap between learned and exact probability paths, improving accuracy without sacrificing efficiency [102].

Despite the theoretical equivalence of the marginals generated by SDEs and their corresponding PF-ODEs, significant practical and theoretical distinctions remain regarding convergence guarantees and likelihood estimation. Recent work has provided non-asymptotic bounds on the Kullback-Leibler divergence for Diffusion Flow Matching models, establishing that convergence depends on moment conditions of the target distribution and the accuracy of the drift approximation [20]. Furthermore, minimax optimality has been demonstrated for probability flow ODE samplers under mild smoothness assumptions, confirming that deterministic sampling can achieve optimal rates in total variation distance without requiring strong structural conditions like log-concavity [181]. However, a critical divergence arises in the context of likelihood training. It has been proven that matching the first-order score function, which is sufficient for generating high-quality samples via SDEs, is inadequate for maximizing the likelihood of the associated ODE [215]. This discrepancy, often termed the "low-noise pathology," necessitates the control of second and third-order score matching errors to ensure optimal likelihood estimation in deterministic settings, as arbitrarily small perturbations in the input can induce large variations in the velocity target when noise levels approach zero [186].

The relationship between these paradigms is further complicated by the geometric properties of the learned vector fields. While diffusion models are conventionally interpreted as learning conservative score functions (gradients of log-densities), empirical evidence suggests that trained neural networks often violate the integral and differential constraints required of true conservative fields [205]. This observation has led to a reinterpretation of diffusion training as flow matching toward the velocity field of a Wasserstein Gradient Flow, where the success of generative sampling does not strictly depend on the vector field being a true score function. Additionally, the gap between ODE and SDE induced distributions can be quantified via the Fokker-Planck residual, with regularization techniques proposed to align the deterministic and stochastic dynamics more closely [228]. Ultimately, the synthesis of these views reveals that FM provides a broader theoretical container, where diffusion models represent a specific stochastic realization, and the PF-ODE serves as the deterministic bridge enabling exact likelihood computation and efficient, high-order sampling [43]. Convergence analysis further supports this by bounding the total variation distance between target and generated distributions based on score matching errors and discretization steps, validating the robustness of deterministic solvers under mild assumptions [269].

Unifying Frameworks and Probability Paths

The theoretical convergence of diffusion and flow paradigms rests on the mathematical equivalence between Stochastic Differential Equations (SDEs) and Probability Flow Ordinary Differential Equations (PF-ODEs), where generative modeling is fundamentally framed as learning a time-dependent vector field that transports a source distribution to a data distribution. A unified measure-theoretic view demonstrates that diffusion models, score-based generative models, and flow matching are all instances of learning a vector field that induces a family of marginals governed by a continuity or Fokker-Planck equation [43]. Within this framework, the reverse-time sampling inherent in diffusion models can be derived as controlled stochastic dynamics, while the PF-ODE yields identical marginals, thereby connecting stochastic diffusion processes to likelihood-based normalizing flows [43]. This connection is further solidified by the observation that the PF-ODE allows for exact likelihood computation and improved sampling efficiency, encapsulating previous approaches in score-based modeling under a single SDE framework [115]. Flow Matching (FM) extends this unification by introducing a simulation-free approach for training Continuous Normalizing Flows (CNFs) based on regressing vector fields of fixed conditional probability paths [13]. Unlike traditional diffusion paths, FM is compatible with a general family of Gaussian probability paths, which subsumes existing diffusion paths as specific instances while allowing for more efficient trajectories, such as those defined by Optimal Transport (OT) displacement interpolation [13].

The geometric interpretation of these processes reveals that both the forward noising and backward generative processes in score-based models can be understood as Wasserstein gradient flows in the space of probability measures [155]. This geometric perspective is critical because it clarifies the nature of the learned vector fields. Contrary to the common assumption that diffusion models strictly learn the score function (the gradient of the log-density), numerical evidence suggests that trained networks often violate the integral and differential constraints required of true conservative vector fields [205]. Instead, diffusion training is more accurately interpreted as flow matching to the velocity field of a Wasserstein Gradient Flow (WGF). Under this WGF perspective, the "probability flow" arises naturally without invoking reverse-time SDE theory, explaining why generative sampling remains successful even when the neural vector field is not a true score function [205]. This reframing eliminates the paradox of non-conservative errors in neural approximations, showing that such errors do not necessarily harm the density transport required for high-quality generation. Furthermore, the gap between ODE and SDE-induced distributions in score-based models can be rigorously described and bounded using the Fokker-Planck equation, with regularization techniques proposed to close this gap and improve ODE sample quality [228].

These OT-based paths are theoretically grounded in minimizing the transport cost between source and target distributions, resulting in straighter trajectories that reduce the global truncation error of numerical ODE solvers and enable faster sampling [9]. Theoretical guarantees have been established for Diffusion Flow Matching (DFM) models, providing non-asymptotic bounds on the Kullback-Leibler divergence between the target and generated distributions under mild assumptions on the score and drift approximation errors [20]. Recent work has derived minimax optimality guarantees for the probability flow ODE, demonstrating that deterministic samplers can achieve near-optimal rates in total variation distance without requiring strong structural conditions on the target distribution [181]. Empirical analyses indicate that FM learns highly rectified transport paths with near-optimal curvature, whereas standard diffusion trajectories remain stochastic and tortuous, leading to an "efficiency frontier" where FM retains high fidelity at significantly fewer function evaluations [136]. However, this framework suffers from a fundamental instability in the low-noise regime, where arbitrarily small perturbations can induce large variations in the velocity target, necessitating hybrid training protocols like Local Contrastive Flow to stabilize representation quality [186].

To address inherent limitations in standard formulations, recent advancements propose learning global state transitions directly rather than local velocity fields. Methods like Transition Flow Matching enable few-step or one-step generation without relying on distillation from pre-trained teachers by modeling the generation trajectory itself as a global quantity [92]. Additionally, the inherent heterogeneity of source and target distributions can pose singularity problems for uniform ODEs, which are effectively tackled by Switched Flow Matching frameworks that eliminate singularities via switching ODEs [132]. The integration of normalizing flows with diffusion processes, such as in Diffusion Normalizing Flow, combines the ability to learn distributions with sharp boundaries from flows with the robustness of diffusion, requiring fewer discretization steps for sampling [61]. The unification also extends to infinite-dimensional function

spaces, where analogs of the probability-flow ODE have been derived to accelerate inference in function generation tasks [236]. The relationship between these paradigms is also explored through the lens of control theory, where the Chow-Rashevskii theorem motivates parameter-efficient generative models that construct expressive flows from a small set of fixed vector fields modulated by learned scalar controls [46]. Ultimately, the synthesis of these approaches suggests that the distinction between diffusion and flow models is increasingly blurred, with both converging toward a unified framework of learning time-dependent vector fields that govern the evolution of probability densities via continuity equations [19].

Trajectory Rectification and Straightening Mechanisms

The transition from stochastic, curved diffusion trajectories to deterministic, straight paths represents a fundamental paradigm shift aimed at minimizing global truncation error and resolving the inference bottleneck of standard Denoising Diffusion Probabilistic Models (DDPMs). While DDPMs often require up to 1,000 iterative steps due to the tortuous nature of their Stochastic Differential Equation (SDE) trajectories, Flow Matching (FM) and Rectified Flow paradigms learn velocity fields that define near-linear Ordinary Differential Equations (ODEs), enabling high-fidelity generation with drastically fewer function evaluations [136]. The core geometric motivation is that rectified flows learn highly straightened, near-optimal transport paths, allowing lightweight Euler solvers to suffice where high-order solvers were previously necessary to manage the curvature-induced errors of diffusion models [136]. This straightening directly reduces the discretization error inherent in numerical integration, which is the primary cause of quality degradation in low-step regimes [9].

Achieving this straightness fundamentally relies on optimizing the coupling between the source noise and target data distributions. Traditional flow matching often employs independent couplings, which result in intersecting trajectories and curved vector fields due to trajectory superposition [208]. To address this, Rectified Flow introduces an iterative "reflow" procedure that progressively straightens the probability path by updating the coupling based on generated samples [9]. However, this iterative approach can be computationally intensive and prone to error accumulation or model collapse if not carefully managed [49]. Consequently, recent research has pivoted toward single-stage or more efficient mechanisms to enforce straightness without extensive retraining. For instance, Isokinetic Flow Matching (Iso-FM) introduces a lightweight dynamical regularizer that directly penalizes pathwise acceleration using a finite-difference approximation of the material derivative, effectively enforcing local velocity consistency without expensive second-order autodifferentiation [208]. Similarly, Optimal Acceleration Transport (OAT-FM) bridges flow matching with optimal acceleration theory, demonstrating that minimizing physical action associated with acceleration is mathematically equivalent to the straightening objective, thus providing a principled condition for flow straightness that can be applied as a fine-tuning stage [117].

Alternative strategies seek to bypass the limitations of iterative reflow by modifying the training objective or the interpolant structure itself. MeanFlow attempts to model the average velocity over time to enable one-step generation but often suffers from slow convergence when trained on highly curved flows [225]. To mitigate this, Rectified MeanFlow combines mean velocity modeling with a single reflow step and a truncation heuristic to reduce residual curvature, achieving superior efficiency and sample quality [225]. Furthermore, Consistency Flow Matching explicitly enforces self-consistency in the velocity field across different times to define straight flows directly, converging significantly faster than both consistency models and standard rectified flow approaches [126]. Another innovative approach involves learning flexible, potentially curved interpolants to ultimately yield straight vector fields, formulating the problem as a multi-level optimization that avoids the rigidity of linear interpolants [3]. Additionally, Transition Flow Matching proposes a unified framework that directly models global state transitions rather than local velocity fields, enabling arbitrary step sizes while maintaining theoretical connections to mean velocity models [92].

The role of Optimal Transport (OT) is central to these mechanisms, as OT provides the theoretical foundation for non-crossing, straight trajectories. However, computing exact OT couplings in high dimensions is intractable. Methods like Cluster-wise Optimal Transport Flow Matching (COT-FM) address this by partitioning data into clusters and assigning dedicated source distributions, effectively reducing the global OT problem to manageable local matches that yield straighter vector fields [78]. Similarly, designing optimal transport flows by treating the prior as a design choice—specifically using low-frequency projections of images—allows for an identity coupling that is empirically OT-optimal, reducing trajectory curvature by more than twofold compared to existing methods [166]. Despite these advances, contradictions exist regarding the necessity of perfect straightness versus the quality of the learned field; for example, while Direct Consistency Models reduce ODE solving error, they sometimes result in worse sample quality, suggesting that strict adherence to the ODE solution is not the sole determinant of generative fidelity [286]. This finding highlights a critical nuance: minimizing numerical error does not automatically translate to perceptual improvement if the underlying manifold structure is compromised.

Moreover, the integration of straightening mechanisms with guidance techniques presents unique challenges. Naive application of Classifier-Free Guidance (CFG) to rectified flows can cause off-manifold drift and visual artifacts

because linear extrapolation of velocities breaks the geometry of the learned straight paths. Rectified-CFG++ resolves this by employing an adaptive predictor-corrector scheme that anchors samples near the learned transport path while applying geometry-aware conditioning, ensuring stability across varying guidance strengths [88]. Additionally, Momentum Guidance offers a plug-and-play alternative that leverages the ODE trajectory itself, using an exponential moving average of past velocities to sharpen samples without the computational overhead of unconditional branches [64]. These developments highlight that while minimizing curvature is essential for efficiency, maintaining the geometric integrity of the data manifold during guided sampling is equally critical for high-quality generation. Ultimately, the convergence of diffusion and flow paradigms hinges on these rectification mechanisms, which transform stochastic, inefficient processes into deterministic, rapid generators capable of real-time application [43].

Theoretical Challenges and Pathologies

The theoretical convergence of diffusion and flow paradigms is frequently obstructed by intrinsic instabilities that arise when attempting to reconcile deterministic efficiency with stochastic robustness. While a unified measure-theoretic framework establishes that diffusion models, score-based generative models, and flow matching are all instances of learning time-dependent vector fields governed by continuity or Fokker-Planck equations, practical implementations reveal significant divergences [43]. A primary theoretical limitation identified in recent literature is the "low-noise pathology," a phenomenon where the optimization landscape becomes increasingly ill-conditioned as noise levels approach zero [186]. In this regime, arbitrarily small perturbations in the input can induce disproportionately large variations in the velocity target, causing the condition number of the learning problem to diverge [186]. This ill-conditioning forces the encoder to reallocate its limited Jacobian capacity toward noise directions rather than semantic features, thereby degrading representation quality [186]. This issue is compounded by the failure of unconstrained neural networks to satisfy necessary boundary conditions, leading to inaccurate velocity field estimations that deviate from the desired Ordinary Differential Equation (ODE) near the data manifold [191].

This tension is further exemplified by the persistent performance gap between ODE-based deterministic samplers and Stochastic Differential Equation (SDE)-based stochastic editors. While ODE solvers offer superior sampling efficiency and exact likelihood computation, empirical evidence consistently suggests that SDE-based methods yield higher fidelity in image editing tasks [232]. Theoretical analysis reveals that the Kullback-Leibler divergence between the marginal distributions of editing SDEs decreases over time, whereas the divergence for ODEs remains static as time approaches zero, indicating a fundamental limitation in the expressivity of deterministic trajectories for editing purposes [232]. To address this discrepancy, recent work has turned to the Fokker-Planck equation to rigorously characterize the divergence between the two induced distributions [228]. By deriving a theoretical upper bound on the Wasserstein 2-distance in terms of a Fokker-Planck residual, researchers have demonstrated that conventional score-based models exhibit significant differences between their deterministic and stochastic counterparts [228]. A promising strategy involves incorporating the Fokker-Planck residual as a regularization term during training; however, this often improves the ODE distribution at the cost of degraded sample quality for the SDE, highlighting a difficult trade-off where optimizing for one paradigm may inadvertently harm the other [228].

Furthermore, the challenge of trajectory curvature poses a significant barrier to efficient ODE sampling, as high curvature leads to substantial global truncation errors, particularly in low-number-of-function-evaluations regimes [9]. Methods such as Rectified Flow aim to straighten these trajectories, yet they often face issues related to singularity and the inability to transport between simple distributions due to the existence and uniqueness constraints of initial value problems in ODEs [132]. The introduction of switched ODEs offers a potential resolution by allowing the selection of specific vector fields based on the heterogeneity of the source and target distributions, thereby eliminating singularities that plague uniform ODE formulations [132]. Additionally, the integration of stochastic starts in ODE samplers for diffusion bridge models has been shown to overcome singular behavior at the beginning of the reverse process, ensuring a smooth transition from corrupted images to the generative trajectory [249].

The interplay between reversibility and numerical stability also presents a theoretical hurdle, particularly in image editing applications. While algebraically reversible ODE solvers eliminate inversion errors, they often exhibit instabilities when subjected to large semantic changes, leading to a trade-off between background preservation and prompt alignment [267]. This instability suggests that exact reversibility may be too rigid for practical editing tasks, prompting the exploration of near-reversible methods that incorporate vector-field smoothing to maintain stability under large edits [267]. Moreover, the assumption that learned vector fields in diffusion models represent true conservative score functions has been challenged; numerical evidence indicates that trained networks often violate the integral and differential constraints required of conservative fields [205]. This suggests that diffusion training is better understood as flow matching to the velocity field of a Wasserstein Gradient Flow rather than strict score learning, providing a theoretical basis for why generative sampling remains successful even when the neural vector field is not a true score [205]. Ultimately, resolving these pathologies requires a nuanced understanding of how stochasticity, curvature, and boundary conditions interact within the continuous-time dynamics of generative models.

Architectural Innovations and Scalability

The architectural landscape of generative modeling has undergone a paradigmatic shift, moving decisively away from traditional U-Net backbones toward Transformer-based architectures and the integration of invertible attention mechanisms. This evolution addresses historical limitations in flow-based models, particularly their struggle with modeling long-range dependencies and scaling to high resolutions due to bijectivity constraints. Early iterations of normalizing flows were often restricted in width and expressivity because the requirement for invertible transformations limited the complexity of individual layers. To mitigate this, researchers introduced densely connected architectures that fuse blocks with Nystro?m self-attention, significantly enhancing density estimation capabilities under moderate computing budgets [123]. Concurrently, the development of invertible attention mechanisms, utilizing masked schemes to ensure tractable Jacobian determinants, has allowed for the seamless integration of attention modules within flow-based models, enabling them to capture long-term data dependencies more efficiently than previous convolutional approaches [38]. Further refinements include the introduction of contractive flows with Lipschitz-constrained self-attention, which improve representation power and convergence speed while maintaining robustness against noise perturbations [203].

A pivotal advancement in this domain is the resurgence of Normalizing Flows through the adoption of Autoregressive Transformer architectures, challenging the notion that diffusion models are inherently superior for high-fidelity synthesis. The TARFlow architecture, which stacks autoregressive Transformer blocks on image patches, has demonstrated that Normalizing Flows can achieve state-of-the-art likelihood estimation and sample quality comparable to diffusion models [1]. Building on this, STARFlow introduces a scalable "deep-shallow" design where a deep Transformer block captures the majority of the model's capacity, followed by computationally efficient shallow blocks. By operating in the latent space of pretrained autoencoders, STARFlow achieves high-resolution image synthesis with exact maximum likelihood training, marking a significant milestone in scaling flow models to resolutions previously dominated by diffusion [7]. Complementing these architectural shifts, the emergence of State Space Models (SSMs), such as Mamba, offers a linear-complexity alternative to the quadratic scaling of Transformers. Architectures like SISMA leverage Mamba blocks for semantic image synthesis, achieving superior efficiency and speed while maintaining competitive quality, thereby providing a lightweight substitute for heavy Transformer-based diffusion models [250]. Similarly, the application of Mamba in medical imaging via the CLIMB framework demonstrates substantial reductions in computational overhead for modeling temporal brain structure evolution compared to self-attention-based latent diffusion models [248].

Parallel to architectural diversification, the theoretical framework of Flow Matching has emerged as a robust alternative to Denoising Diffusion Probabilistic Models, offering distinct advantages in efficiency and trajectory geometry that directly influence architectural design choices. Comparative analyses reveal that FM learns highly rectified transport paths with near-optimal curvature, whereas diffusion trajectories remain stochastic and tortuous [136]. This geometric efficiency allows FM to retain high fidelity with significantly fewer function evaluations, making it superior for deployment on low-resource hardware. The flexibility of FM extends to its ability to define probability paths using Optimal Transport displacement interpolation, which yields straighter trajectories and faster sampling compared to standard diffusion paths [13]. However, the choice between Diffusion Bridge and FM remains nuanced; while Diffusion Bridge may offer lower cost functions and more stable trajectories under Stochastic Optimal Control, FM can suffer from ineffective interpolation coefficients when training data is scarce, necessitating careful selection based on the specific task and data availability [239].

To further enhance scalability and efficiency, recent innovations have focused on accelerating inference through one-step or few-step generation strategies, often leveraging hybrid architectures. Transition Flow Matching proposes a paradigm that directly learns global state transitions rather than local velocity fields, enabling generation in arbitrary step sizes without relying on distillation from pretrained teachers [92]. Similarly, Flow Generator Matching accelerates sampling to a single step while maintaining performance, successfully distilling large-scale text-to-image models into efficient one-step generators [213]. The integration of these efficient flow models with latent space modeling has also proven critical; for instance, boosting latent diffusion with FM allows small diffusion models to provide visual diversity while flow matching efficiently maps latents to high-resolution images, achieving synthesis at high resolutions with minimal computational cost [182]. Furthermore, the development of Normalizing Trajectory Models bridges the gap

between diffusion and flows by modeling reverse conditionals as conditional normalizing flows, retaining exact likelihood training while achieving high-quality generation in as few as four steps [197]. These architectural and algorithmic synergies collectively demonstrate that the integration of Transformers, State Space Models, and advanced flow matching techniques is redefining the boundaries of scalability, efficiency, and fidelity in generative modeling.

Transformer-Based and Attention-Enhanced Flows

The integration of Transformer architectures into flow-based generative models has fundamentally reshaped the landscape of likelihood-based synthesis, effectively merging the scalability of autoregressive modeling with the continuous representational power of normalizing flows. A pivotal advancement in this domain is the Transformer Autoregressive Flow (TARFlow), which utilizes stacks of autoregressive Transformer blocks on image patches to model continuous distributions. This approach has been theoretically proven to possess universality for modeling continuous distributions and has set new benchmarks in likelihood estimation, surpassing previous methods by significant margins while generating samples comparable to diffusion models [1]. Building upon this foundation, STARFLOW introduces scalable architectural innovations designed to handle high-resolution image synthesis within the latent space of pretrained autoencoders. A key contribution of STARFLOW is its deep-shallow design, where a deep Transformer block captures the majority of the model's capacity, followed by computationally efficient shallow blocks that refine spatial details without incurring prohibitive costs [7]. While STARFLOW focuses on architectural depth distribution, further iterations such as iterative TARFlow (iTARFlow) incorporate diffusion-style iterative denoising procedures during sampling while maintaining a fully end-to-end, likelihood-based training objective, thereby advancing the frontier of flow models on large-scale datasets like ImageNet [17]. These divergent strategies highlight a critical tension in the field: balancing the exact likelihood guarantees of pure flows with the sampling efficiency and quality gains borrowed from diffusion paradigms.

Beyond autoregressive structures, the direct incorporation of attention mechanisms into invertible layers has addressed historical limitations regarding long-range dependency modeling in flows. Traditional flow layers often struggle with expressivity due to the constraints imposed by invertibility, but Attentive Contractive Flow (ACF) resolves this by utilizing localized self-attention within a contractive flow framework. ACF enforces Lipschitz constraints to maintain invertibility while capturing global context, resulting in improved bits per dimension metrics and enhanced resilience to additive noise compared to standard flow models [203]. In contrast to the contractive approach, Generative Flows with Invertible Attentions propose both map-based and transformer-based attention mechanisms that employ masked schemes to ensure tractable Jacobian determinants. These mechanisms allow for the seamless integration of attention modules at any position within flow-based models, significantly improving efficiency in modeling long-term data dependencies for both unconditional and conditional tasks [38]. The synergy between Transformers and flows extends to volumetric data as well; VolDiT introduces the first purely transformer-based 3D Diffusion Transformer for medical image synthesis. By replacing convolutional U-Net backbones with global self-attention over 3D patch embeddings, VolDiT achieves superior global coherence and anatomical consistency in volumetric generation, demonstrating the versatility of transformer-based backbones in high-dimensional spaces [204].

The convergence of diffusion models and normalizing flows has also led to hybrid architectures that leverage the strengths of both paradigms, often utilizing these advanced transformer backbones to enhance performance. DiNof (Diffusion with Normalizing flow priors) utilizes normalizing flows to parameterize noisy data at arbitrary diffusion steps, serving as a deterministic prior that reduces the total number of required diffusion steps and accelerates both forward and backward processes [53]. In the realm of inverse problems, FlowDPS extends diffusion inverse solvers into the flow framework by decomposing the flow ordinary differential equation into components for clean image and noise estimation. This solver can be seamlessly integrated into latent flow models with transformer architectures, outperforming state-of-the-art alternatives in linear inverse problems without requiring additional training [240]. Furthermore, the application of Transformers in flow matching has enabled efficient one-step sampling and robust performance in specialized domains. For instance, MultiCE-Flow employs a Diffusion Transformer (DiT) backbone guided by multimodal environmental data to reconstruct high-fidelity channels in communication systems, leveraging flow matching to learn linear trajectories for efficient sampling [254].

Architectural refinements continue to optimize the trade-off between computational cost and generation fidelity within these transformer-enhanced frameworks. RelaCtrl addresses the inefficiency of controlled Diffusion Transformers by evaluating the relevance of control information across different transformer layers, tailoring the positioning and capacity of control layers to reduce unnecessary parameters while maintaining generation quality [270]. Additionally, Laplacian multiscale flow matching (LapFlow) enhances flow matching by processing multi-scale representations in parallel through a mixture-of-transformers architecture with causal attention, eliminating the need for explicit renoising

between scales and accelerating sampling for high-resolution generation [52]. Theoretical analyses further support these architectural choices, with studies indicating that bi-Lipschitz normalizing flows, when viewed through the lens of score-based diffusion, possess universal distributional approximation capabilities, ensuring that Gaussian pullbacks induced by such maps are dense among all probability densities [178]. Collectively, these advancements illustrate that the fusion of Transformer architectures, attention mechanisms, and flow dynamics creates a robust framework capable of handling complex, high-dimensional data distributions with unprecedented efficiency and quality.

Latent Space Modeling and Efficient Backbones

The strategic pivot from direct pixel-space manipulation to operations within compressed latent representations marks a fundamental maturation in generative modeling, driven by the necessity to scale to higher resolutions while maintaining computational feasibility. Operating in a lower-dimensional manifold significantly reduces the complexity of the generation process, yet early approaches utilizing normalizing flows often struggled with capacity constraints due to strict invertibility requirements that limited model width. To address these limitations, recent architectures have integrated dense connectivity and self-attention mechanisms to enhance expressivity without sacrificing the benefits of flow-based modeling [123]. Furthermore, the introduction of noisy injective flows has allowed these models to transcend dimensionality constraints, mapping latent spaces to learnable manifolds in high-dimensional data spaces, thereby improving sample quality and embedding separability [73]. The efficacy of this latent-centric paradigm is further evidenced by works that combine variational autoencoders with diffusion processes, equipping models with low-dimensional, interpretable latent codes that facilitate controllable synthesis and improve the speed-quality trade-off [DiffuseVAE]. By learning in the latent space of pretrained autoencoders, models like STARFlow demonstrate that this approach is far more effective than modeling pixels directly, achieving competitive results in high-resolution synthesis [7].

A critical bottleneck in scaling latent space models, particularly those based on transformers, has been the quadratic complexity of attention mechanisms, which becomes prohibitive at high resolutions. In response, the field has witnessed a significant adoption of linear-complexity backbones, most notably State Space Models (SSMs) such as Mamba. These architectures offer a compelling alternative to transformers by maintaining linear memory growth with respect to sequence length, enabling efficient semantic synthesis in resource-constrained environments. The application of Mamba to Semantic Image Synthesis (SIS) exemplifies this shift; the SISMA architecture replaces traditional transformer blocks with Mamba-based units, achieving superior Fr chet Inception Distance (FID) scores while operating at significantly increased speeds compared to state-of-the-art diffusion models [250]. Similarly, in the medical imaging domain, the CLIMB framework leverages Mamba to model temporal changes in brain structure, substantially reducing computational overhead compared to self-attention-based latent diffusion models while preserving high-fidelity synthesis [248]. These developments highlight a consensus that linear-complexity backbones are essential for the next generation of scalable generative systems, effectively decoupling semantic richness from computational cost. This efficiency is further corroborated by Laplacian multiscale flow matching approaches that process different scales in parallel to accelerate sampling while maintaining quality [52].

Parallel to architectural innovations in backbones, the methodology for traversing latent spaces has evolved from stochastic diffusion processes to deterministic flow matching. Flow matching offers a more direct optimization objective by learning rectified transport paths between noise and data distributions, resulting in smoother trajectories and faster convergence. Comparative analyses indicate that flow matching learns near-optimal geometric paths, whereas traditional diffusion trajectories remain stochastic and tortuous, leading to an "efficiency frontier" where flow matching retains high fidelity with significantly fewer function evaluations [136]. This efficiency is crucial for high-resolution synthesis, as demonstrated by methods that boost latent diffusion with flow matching to achieve state-of-the-art results at resolutions up to 2048^2 pixels with minimal computational cost [182]. Moreover, flow matching has proven effective in specialized domains such as layout generation, where the LayoutFlow model utilizes smooth, directed flows to arrange elements more naturally and rapidly than noisy diffusion counterparts [125]. However, this transition is not without challenges; recent theoretical work identifies a "low-noise pathology" in flow matching where arbitrarily small perturbations can induce large variations in velocity targets, necessitating hybrid training protocols like Local Contrastive Flow to stabilize representation quality [186].

The integration of efficient backbones and advanced latent traversal techniques has enabled more sophisticated control mechanisms that operate with greater precision and flexibility. By structuring the latent space into multi-level clusters, models can perform customized truncation and semantic control at coarse, medium, and fine levels, offering a better trade-off between diversity and realism [299]. Additionally, energy-based models operating in the latent space have facilitated compositional generation, allowing for the zero-shot combination of unseen attributes through the composition of energy functions [158]. The synergy between latent space modeling and efficient backbones is further reinforced by the development of guidance strategies that operate directly on the flow map, such as Flow Map Reward

Guidance, which achieves alignment with human preferences using single-trajectory inference and drastically reduced sampling steps [148]. Furthermore, rethinking control as a feedback system has led to architectures like ControlNet-XS, which optimize the bandwidth and frequency of communication between the controller and the generator to improve fidelity without the heavy parameter overhead of previous methods [162]. Collectively, these advancements underscore a unified trajectory in generative AI: moving towards architectures that decouple semantic richness from computational cost through intelligent latent representations and linear-scaling dynamics.

Hybrid and Free-Form Architectures

The evolution of normalizing flows has been characterized by a persistent tension between the mathematical rigor of exact likelihood estimation and the architectural flexibility required for high-fidelity generation. Historically, the requirement for analytical invertibility constrained flow architectures to specialized coupling layers, limiting their expressiveness compared to diffusion models. A paradigm shift occurred with the introduction of free-form approaches that decouple the generative objective from explicit invertibility. By employing an efficient gradient estimator for the change-of-variables formula, it became possible to train any dimension-preserving neural network as a generative model via maximum likelihood [31]. This relaxation allows researchers to tailor inductive biases directly to the data modality, such as utilizing equivariant networks for molecule generation or standard ResNets for inverse problems, without being restricted to specialized flow layers. Complementing this, bidirectional frameworks have emerged that remove the need for an exact analytic inverse entirely by learning a separate reverse model to approximate the noise-to-data mapping, thereby accelerating sampling by up to two orders of magnitude while maintaining competitive generation quality [138]. These developments collectively demonstrate that strict bijectivity is not a prerequisite for effective likelihood-based training, opening the door to more expressive and task-specific architectures.

Parallel to relaxing invertibility constraints, a robust line of research has focused on hybridizing flows with iterative denoising procedures inspired by diffusion dynamics to bridge the sample quality gap. The "Diffusion Normalizing Flow" introduces a framework based on Stochastic Differential Equations (SDEs) that jointly trains forward and backward neural SDEs, effectively combining the sharp boundary learning capabilities of flows with the sampling efficiency of diffusion processes [61]. Taking this synthesis further, recent work advances the state of normalizing flow generative models by introducing iterative variants that maintain a fully end-to-end, likelihood-based objective during training while employing an autoregressive generation followed by an iterative denoising step during sampling [17]. This hybrid strategy allows models to achieve competitive performance on high-resolution image tasks, challenging the notion that flows must rely solely on single-pass transformations. Another variation utilizes normalizing flows to parameterize noisy data at arbitrary diffusion steps, serving as a deterministic prior that reduces the total number of required diffusion steps while improving expressive power through a combination of deterministic and stochastic mappings [53]. These approaches illustrate a spectrum of integration where iterative refinement is used to correct the limitations of single-step flows without abandoning probabilistic rigor.

Theoretical unification of these paradigms is provided by work establishing a formal equivalence between score-difference flows and denoising diffusion models, offering a unified link between methods addressing the generative modeling trilemma of quality, coverage, and speed [210]. Addressing the specific challenge of few-step generation, new frameworks model each reverse step as an expressive conditional normalizing flow, retaining exact likelihood training while matching the quality of strong baselines in as few as four steps [197]. This is particularly significant as it overcomes the Gaussian assumption bottleneck inherent in standard diffusion transitions when the number of steps is compressed. Furthermore, surveys of recent works highlight a trend toward loosening bijectivity constraints by combining aspects of variational autoencoders and diffusion to balance expressivity and likelihood tractability [288]. To address the inability of standard flows to learn low-dimensional manifold representations, noisy injective flows have been introduced to explicitly map latent spaces to learnable manifolds while accounting for deviations via additive noise [73].

Scalability in these architectures has been significantly advanced through the integration of transformers and novel training objectives. Scalable architectures utilizing autoregressive Transformer blocks on image patches have achieved state-of-the-art likelihood estimation and sample quality comparable to diffusion models when combined with noise augmentation and guidance [1]. Building on this, theoretical universality has been established for such transformer-based flows, introducing deep-shallow designs that enable the successful scaling of normalizing flows to high-resolution image synthesis [7]. The efficiency of these models is further enhanced by simulation-free approaches for training Continuous Normalizing Flows (CNFs) by regressing vector fields of fixed conditional probability paths, subsuming diffusion paths and enabling the use of optimal transport paths for faster convergence [13]. Comparative analyses confirm that flow matching learns near-optimal rectified transport paths that significantly outperform diffusion in efficiency, particularly on low-resource hardware [136]. However, challenges remain in the low-noise regime, where ill-conditioning issues have been identified, prompting proposals for hybrid contrastive training protocols to stabilize

representation quality [186]. Finally, the flexibility of these architectures extends to specialized domains, leveraging specific backbones within flow matching frameworks to predict physical fields on irregular geometries without pixelation artifacts [229]. These advancements collectively signify a shift toward architectures that prioritize both mathematical rigor and practical flexibility, enabling high-fidelity generation across diverse data modalities.

Optimization Strategies and Sampling Efficiency

The primary impediment to the widespread deployment of continuous-time generative models is the computational latency inherent in their sampling procedures, which traditionally require solving high-dimensional Ordinary or Stochastic Differential Equations (ODEs/SDEs) with a large number of function evaluations (NFE) [251]. The core mathematical challenge driving this inefficiency is the global truncation error of numerical solvers, which is exacerbated by the high curvature of probability flow trajectories [9]. When trajectories are curved, standard solvers such as Euler or Runge-Kutta methods accumulate significant discretization errors, particularly in the low-NFE regime required for real-time applications [9]. Consequently, a dominant paradigm for acceleration involves "straightening" these probability flows to minimize curvature, thereby reducing truncation error and enabling high-fidelity generation in fewer steps. Theoretical analyses support these efforts by providing convergence guarantees for probability flow ODEs and establishing error bounds that depend on both score matching accuracy and discretization step size [269].

Training-based acceleration techniques explicitly modify the learning objective or data coupling strategy to enforce straighter paths. Rectified Flow and its variants operate on the principle that linear interpolation between noise and data distributions yields straighter trajectories than the stochastic paths typical of standard diffusion models [9]. However, achieving near-perfect straightness often requires iterative reflow procedures, where the model is retrained on pairs generated by the previous iteration's ODE solver, a process that can be computationally expensive and prone to distribution drift [49]. To mitigate these issues, recent works have introduced improvements such as Sequential Reflow (SEQRF), which systematically straightens the probability flow to reduce global truncation error, and Simple ReFlow, which proposes specific improvements to training dynamics to prevent sample quality deterioration [9; 272]. Furthermore, methods like Balanced Conic Rectified Flow address the drift problem by supervising the model with inversions of real data points rather than relying solely on generated samples, ensuring the ODE paths remain anchored to the true data distribution [49].

Beyond iterative reflow, alternative training strategies aim to learn straight flows in a single stage or through novel architectural constraints. Isokinetic Flow Matching (Iso-FM) introduces a lightweight, Jacobian-free dynamical regularizer that directly penalizes pathwise acceleration, enforcing local velocity consistency without the need for expensive second-order autodifferentiation [208]. Similarly, Block Matching leverages label information to partition the data distribution, matching blocks with a prior to learn straighter flows while controlling the variance to balance diversity and numerical error [104]. Other approaches, such as Learning Straight Flows by Learning Curved Interpolants, propose a counter-intuitive strategy of learning flexible, potentially curved interpolants to ultimately derive straight vector fields, formulated as a multi-level optimization problem [3]. Additionally, Constant Acceleration Flow (CAF) extends the constant velocity assumption of rectified flow by introducing acceleration as a learnable variable, allowing for more expressive estimation of the ODE flow and improved few-step coupling preservation [253].

In contrast to training-based methods that alter the model or data distribution, a parallel line of research focuses on training-free solver optimizations and bespoke numerical integrators tailored to specific pre-trained models. Bespoke solvers represent a novel framework where custom ODE solvers are constructed for a given pre-trained flow model, optimizing a parameter-efficient solver with minimal learnable parameters to significantly improve approximation quality compared to generic dedicated solvers [251]. This approach avoids the costly retraining associated with distillation while achieving high fidelity at low NFEs. Similarly, the Bi-Anchor Interpolation Solver (BA-solver) bridges the gap between training-free versatility and training-based performance by introducing a lightweight SideNet to approximate intermediate velocities, enabling large interval sizes with minimized error without retraining the heavy backbone [230]. Furthermore, DPM-Solver-v3 improves diffusion ODE solving by introducing empirical model statistics to minimize first-order discretization error, incorporating multistep methods and predictor-corrector frameworks to enhance quality at small NFEs [261].

Despite the success of straightening trajectories, recent findings highlight a complex trade-off between trajectory straightness, ODE solving accuracy, and final sample quality. Intriguingly, research into Consistency Models (CMs) reveals that directly minimizing the ODE solving error does not necessarily translate to better generated samples [286]. Direct CMs, which explicitly optimize for ODE solution accuracy, were found to produce significantly worse sample quality compared to standard CMs, suggesting that the success of consistency models may rely on factors beyond mere trajectory fidelity [286]. This contradiction underscores the nuance required in optimization strategies; simply forcing a

straight path or minimizing numerical error may inadvertently compromise the model's ability to capture the full complexity of the data distribution. Moreover, guidance techniques, which are essential for high-quality conditional generation, can interact unpredictably with straightened flows. While Classifier-Free Guidance (CFG) is standard, it doubles inference costs and can reduce diversity. Novel approaches like Momentum Guidance leverage the ODE trajectory itself to provide a sharpening effect without extra computations, while Adaptive Manifold Guidance addresses the structural blow-up of divergence terms near the data manifold to prevent hallucinations [64; 12].

The landscape of optimization also includes distillation techniques that compress multi-step processes into one or few steps. Methods like Trajectory Consistency Distillation (TCD) and Flow Generator Matching (FGM) aim to distill teacher models into students capable of one-step generation, though they must navigate the quality-diversity trade-off often inherent in such compression [238; 213]. Meanwhile, approaches like MeanFlow and Rectified MeanFlow attempt to model the average velocity over time to enable one-step generation, though they face challenges with slow convergence when trained on highly curved flows [225; 202]. The tension between these various strategies—whether to straighten the flow via reflow, design a bespoke solver, or distill the trajectory—reflects the ongoing effort to balance computational efficiency with the preservation of distributional coverage and perceptual quality. Ultimately, the choice of strategy depends on the specific constraints of the application, weighing the costs of additional training against the benefits of faster, training-free inference.

Learning Straight Vector Fields and Optimal Transport

The transition from stochastic diffusion processes to deterministic flow-based approaches represents a fundamental shift in generative modeling, driven by the imperative to minimize sampling steps while preserving sample fidelity. At the theoretical core of this evolution lies the continuity equation, which governs the transport of probability mass from a simple source distribution to a complex target distribution [43]. While traditional Flow Matching models typically rely on linear interpolants coupled with independent noise-data pairings, this standard formulation frequently yields curved vector fields that inflate numerical truncation errors during Ordinary Differential Equation (ODE) integration, thereby necessitating a high number of function evaluations to maintain quality [3]. Recent theoretical reinterpretations suggest that diffusion training is effectively flow matching to the velocity field of a Wasserstein Gradient Flow, implying that while non-conservative errors in neural approximations do not necessarily impede density transport, the geometric straightness of the path remains the primary bottleneck for low-step generation [205]. Consequently, the field has converged on Optimal Transport (OT) theory as the mathematical framework for constructing straight, non-crossing trajectories that enable efficient one-step or few-step sampling.

A primary avenue for achieving straighter fields involves refining the coupling mechanism between source and target distributions to approximate the computationally prohibitive exact OT map. While iterative rectification methods have proven effective, they often suffer from error accumulation and potential model collapse due to reliance on self-generated samples over multiple reflow stages [78]. To overcome these limitations, researchers have developed strategies that localize the transport problem or leverage high-quality priors. Cluster-wise Optimal Transport Flow Matching (COT-FM) partitions data into modes and assigns dedicated source distributions to each cluster, significantly reducing path crossings and yielding straighter fields compared to global random couplings [78]. Alternatively, Straighter Flow Matching via a Diffusion-Based Coupling Prior (StraightFM) bypasses iterative reflow entirely by utilizing the Probability Flow ODE of pre-trained diffusion models to generate high-quality pseudo-couplings in a single training round [140]. This approach capitalizes on the fact that well-trained diffusion models inherently learn the global data distribution, providing a tractable coupling strategy that preserves structural integrity without the overhead of solving minibatch OT problems [140]. Furthermore, treating the prior as a design choice rather than a fixed input allows for the identification of low-frequency projections of natural images as a tractable prior that admits an OT-optimal identity coupling, reducing trajectory curvature by more than twofold [166].

Beyond coupling strategies, direct regularization of the vector field dynamics has emerged as a powerful method for enforcing straightness. Optimal Acceleration Transport (OAT) theory formalizes the intuition that minimizing trajectory curvature is mathematically equivalent to minimizing the physical action associated with acceleration [117]. By optimizing acceleration transport in the product space of sample and velocity, OAT-FM provides a necessary and sufficient condition for flow straightness, enabling a two-phase paradigm where pre-trained models can be fine-tuned for rapid generation without distribution drift [117]. Complementing this, Isokinetic Flow Matching (Iso-FM) introduces a lightweight, Jacobian-free regularizer that penalizes pathwise acceleration using a finite-difference approximation of the material derivative, achieving dramatic improvements in few-step generation without requiring auxiliary encoders [208]. Challenging the assumption of constant velocity inherent in many rectified flows, Constant Acceleration Flow (CAF) introduces acceleration as a learnable variable, allowing for more expressive estimation of ODE flows and superior performance in one-step generation scenarios [253]. These dynamic approaches contrast with Transition Flow Matching, which shifts the learning objective from local velocity fields to global transition flows, naturally supporting generation at arbitrary time points and unifying the perspective with mean velocity models [92].

Despite the consensus on the benefits of straight paths, a critical tension exists between trajectory straightness and the expressivity required to capture complex, multi-modal data distributions. Strictly enforcing linear interpolation can lead to ambiguous velocity fields where multiple ground-truth directions exist at the same location, causing standard mean-squared-error losses to average out essential dynamics [106]. To address this multi-modality, Variational Rectified Flow Matching learns and samples from multi-modal flow directions, capturing complex transport dynamics that deterministic straightening methods miss [106]. Similarly, Gaussian Mixture Flow Matching Models predict dynamic Gaussian mixture parameters to capture multi-modal flow velocity distributions, mitigating the oversaturation issues often seen in classifier-free guidance while improving few-step sampling quality [188]. In cases where data heterogeneity creates singularities that a single ODE cannot resolve, Switched Flow Matching eliminates these issues

by switching between multiple ODEs based on the inherent joint heterogeneity of the source and target distributions [132]. These approaches highlight that while straightness is desirable for efficiency, the underlying transport mechanism must remain flexible enough to handle topological complexities.

Architectural innovations further support these straightening objectives by introducing structured and parameter-efficient formulations. Parameter-Efficient Generative Modeling with Controlled Vector Fields utilizes the Chow-Rashevskii theorem to construct expressive flows from a small set of fixed, bracket-generating vector fields modulated by learned scalar controls, offering a geometric alternative to unconstrained high-dimensional learning [46]. In specialized domains such as medical imaging, where data scarcity and structural fidelity are paramount, Medical Optimal Transport Flow Matching (MOTFM) adapts these straight mapping principles to synthesize high-fidelity 2D and 3D images, demonstrating that OT-based straightening preserves anatomical integrity better than traditional diffusion solvers [18]. The convergence of OT theory, dynamic regularization, and structured coupling strategies defines the current frontier in optimizing sampling efficiency, moving the field toward robust, high-quality generation in extremely low-step regimes while navigating the delicate balance between geometric simplicity and distributional expressivity.

Advanced ODE Solvers and Distillation

The computational bottleneck inherent in flow-based generative models stems from the necessity of solving high-dimensional Ordinary Differential Equations (ODEs) through iterative numerical integration, a process that traditionally demands a large number of function evaluations (NFE) to approximate solutions accurately. To address this, recent research has bifurcated into two primary strategies: the development of bespoke, empirically optimized ODE solvers tailored to specific pre-trained models, and the distillation of multi-step processes into efficient one-step or few-step generators. While standard dedicated solvers often require significant NFE to maintain sample fidelity, bespoke approaches optimize solver parameters specifically for the ODE trajectory of a given flow model. For instance, bespoke solvers have demonstrated the ability to achieve state-of-the-art Frechet Inception Distance (FID) scores on CIFAR10 and ImageNet with as few as 10 to 20 NFE, significantly outperforming generic dedicated solvers while requiring only a fraction of the GPU time needed for full model training [251]. Similarly, the Bi-Anchor Interpolation Solver (BA-solver) introduces a lightweight SideNet to approximate bidirectional temporal velocities, enabling high-order integration with large interval sizes and achieving quality comparable to 100+ NFE Euler solvers in just 10 steps [230]. Further enhancing solver efficiency, DPM-Solver-v3 leverages empirical model statistics to minimize first-order discretization errors, offering substantial speed-ups in the low-NFE regime for both pixel-space and latent-space models [261].

Parallel to solver optimization, distillation frameworks aim to compress the generative trajectory itself, effectively transforming the continuous flow into a discrete, rapid mapping. Flow Generator Matching (FGM) represents a significant advancement in this domain, providing theoretical guarantees for accelerating flow-matching models into one-step generation without sacrificing performance. FGM has achieved record-breaking FID scores on CIFAR10 and successfully distilled large-scale text-to-image models like Stable Diffusion 3 into single-step generators that rival multi-step counterparts in quality [213]. Building on the concept of trajectory straightening, Consistency Flow Matching explicitly enforces self-consistency in the velocity field to define straight flows, converging significantly faster than traditional consistency models while improving generation quality [126]. However, the relationship between ODE solving accuracy and sample quality is not always linear; intriguingly, Direct Consistency Models, which directly minimize ODE solving error, have been found to produce significantly worse samples than standard Consistency Models, suggesting that the success of distillation may rely on objectives that diverge from strict ODE fidelity [286].

To facilitate these distillation processes, various techniques focus on "straightening" the probability flow to reduce the curvature of ODE trajectories, thereby minimizing global truncation errors in low-NFE regimes. Sequential Reflow (SEQRF) and Simple Reflow propose iterative learning techniques to straighten these paths, with the latter introducing seven specific improvements to training dynamics that mitigate the sample deterioration often associated with reflow procedures [9] [272]. Complementing these efforts, Constant Acceleration Flow (CAF) challenges the assumption of constant velocity in rectified flows by introducing acceleration as a learnable variable, resulting in more expressive flow estimation and superior one-step generation performance [253]. Furthermore, Straighter Flow Matching utilizes a diffusion-based coupling prior to establish scalable couplings that straighten trajectories without the need for computationally prohibitive optimal transport calculations or multi-round training [140]. Some approaches even propose learning flexible, curved interpolants specifically to induce straighter vector fields, demonstrating that the path definition itself can be optimized for inference efficiency [3].

Advanced distillation paradigms also explore policy-based and variational approaches to bridge the gap between teacher and student models. Policy-based flow models (π -Flow) modify the student output to predict a network-free policy, enabling dynamic velocity prediction at substeps with negligible overhead and avoiding the quality-diversity trade-off common in distillation [187]. Similarly, Normalizing Trajectory Models (NTM) model each reverse step as an expressive conditional normalizing flow, retaining exact likelihood training while achieving high-quality generation in just four steps, a distinct advantage over methods that sacrifice the likelihood framework for speed [197]. In the realm of mean-velocity modeling, α -Flow addresses optimization conflicts between trajectory flow matching and consistency terms through a curriculum strategy, achieving state-of-the-art results with vanilla DiT backbones [202]. Additionally, Transition Flow Matching proposes a unified framework that directly learns global state transitions rather than local vector fields, supporting arbitrary step sizes and enabling end-to-end training from scratch [92].

Despite these advancements, challenges remain regarding the stability and generalizability of few-step methods. For example, while Rectified Flow enables one-step sampling, it often suffers from drift and error accumulation during the reflow process; Balanced Conic Rectified Flow addresses this by supervising the model with inversions of real data points, ensuring ODE paths remain directed toward real samples [49]. Moreover, the application of these techniques extends beyond standard image generation to specialized domains such as medical imaging, where Optimal Transport Flow Matching has been shown to significantly accelerate inference while preserving the diagnostic quality required for tasks like segmentation and synthesis [18]. The integration of reinforcement learning further refines these models, as seen in SuperFlow, which employs variance-aware sampling and step-level advantage estimates to optimize flow-based text-to-image models more efficiently than previous RL baselines [179]. Collectively, these developments illustrate a rapidly evolving landscape where bespoke solvers and sophisticated distillation frameworks are systematically dismantling the computational barriers of flow-based generative modeling.

Dynamic Step Size and Temporal Optimization

The computational bottleneck of flow-based generative models lies in the numerical integration of Ordinary Differential Equations (ODEs) during sampling, where high-curvature trajectories necessitate small step sizes to maintain fidelity. To mitigate this, recent research has shifted from static integration schemes to adaptive strategies that optimize the temporal dynamics of the flow itself. A primary approach involves treating the time discretization schedule as a learnable or optimizable parameter rather than a fixed hyperparameter. The Bellman Optimal Stepsize Straightening technique exemplifies this by employing dynamic programming to determine optimal step sizes for pretrained flow-matching models under strict computational budgets. This method refines the velocity network to match these optimal intervals, effectively "straightening" generation paths to adhere to specific resource constraints while preserving sample quality [127]. Distinct from this post-hoc optimization, Temporal Optimization treats evolutionary time as a learnable parameter integrated directly into the training loop. By alternating between optimizing network weights and evolutionary time coordinates via coordinate descent, this framework discovers a temporal parameterization that accelerates forward propagation without sacrificing the invertibility or stability of the flow [76]. While the former focuses on distilling existing models for low-resource scenarios, the latter reshapes the underlying vector field dynamics during training, highlighting a divergence between adaptive inference strategies and fundamental temporal learning.

Complementing temporal optimization is the explicit regularization of trajectory curvature to facilitate larger integration steps. Isokinetic Flow Matching introduces a lightweight, Jacobian-free dynamical regularizer that directly penalizes pathwise acceleration using a self-guided finite-difference approximation. By enforcing local velocity consistency, this method dramatically reduces global truncation error, enabling high-quality generation with significantly fewer steps [208]. Similarly, Constant Acceleration Flow challenges the assumption of constant velocity inherent in rectified flow models, introducing acceleration as an additional learnable variable to more accurately estimate the ODE flow. This allows for more expressive trajectory modeling that outperforms standard rectification techniques in few-step generation scenarios [253]. These acceleration-based regularizers agree on the premise that minimizing curvature is essential for efficiency, yet they differ in implementation; one operates as a plug-and-play addition, whereas the other modifies the fundamental parameterization of the flow dynamics. Furthermore, Optimal Acceleration Transport bridges flow matching with optimal acceleration theory, demonstrating that straightening objectives are mathematically equivalent to minimizing physical action, and proposes a two-phase paradigm to fine-tune models via acceleration transport without risking data distribution drift [117].

Beyond direct regularization, alternative paradigms have emerged that reframe the generation process to bypass iterative solving or reduce dependency on fine-grained time steps. Transition Flow Matching proposes a shift from learning local velocity fields to modeling global state transitions directly, deriving a Transition Flow Identity that enables generation at arbitrary time points and step sizes from scratch [92]. In a related vein, Normalizing Trajectory Models address the "Gaussian bottleneck" of few-step diffusion by modeling each reverse step as an expressive conditional normalizing flow, capturing non-Gaussian reverse conditionals with exact likelihood training to achieve high fidelity in as few as four steps [197]. The Bi-Anchor Interpolation Solver introduces a hybrid approach where a lightweight SideNet learns to approximate intermediate velocities between high-precision anchors established by a frozen backbone, achieving quality comparable to high-step solvers in just a fraction of the evaluations [230]. Contrasting with these training-intensive methods, Bespoke Solvers construct custom ODE solvers tailored to a pretrained model with minimal training overhead, optimizing a parameter-efficient solver structure to achieve significant gains in approximation quality without retraining the entire generative backbone [251].

The tension between single-stage regularization and multi-stage rectification underscores ongoing debates regarding training complexity versus trajectory straightness. Sequential Reflow argues for an iterative learning technique that progressively straightens the probability flow to reduce global truncation error, suggesting that multi-stage training may be necessary to achieve the highest levels of straightness [9]. Conversely, methods like Simple Reflow examine the design space of iterative procedures to mitigate sample deterioration, proposing improvements that achieve state-of-the-art scores with merely a handful of neural function evaluations [272]. Additionally, the integration of optimal transport theory remains a contentious but fruitful area; while Cluster-wise Optimal Transport Flow Matching utilizes clustering to approximate optimal couplings locally for straighter paths, other works suggest that solving the

global optimal transport problem is computationally prohibitive, favoring instead the learning of flexible interpolants or the use of diffusion-based coupling priors to approximate straight trajectories [78] [140]. Ultimately, the convergence of dynamic step size optimization, temporal learning, and curvature regularization represents a sophisticated evolution in flow matching, moving the field toward real-time, high-fidelity generative capabilities.

Conclusion and Future Directions

The synthesis of diffusion and flow-based generative paradigms has culminated in a unified theoretical framework where both approaches are understood as instances of learning time-dependent vector fields that govern the evolution of probability densities. This convergence, grounded in measure theory and optimal transport, reveals that stochastic diffusion processes and deterministic normalizing flows are not mutually exclusive but rather complementary manifestations of the same underlying continuity equations. By decoupling the generative path from rigid Gaussian constraints, Flow Matching has emerged as a superset that subsumes traditional diffusion models, enabling the construction of arbitrary probability paths that facilitate exact likelihood computation and more efficient sampling trajectories. The realization that score-based models possess deterministic Probability Flow ODE counterparts has bridged the gap between stochastic robustness and deterministic efficiency, allowing researchers to leverage the strengths of both worlds within a single mathematical container.

A central theme driving this unification is the geometric rectification of generative trajectories. The field has shifted focus from merely refining model architectures to optimizing the curvature of the transport paths themselves. Through the lens of Optimal Transport, methods such as Rectified Flow and various straightening mechanisms have demonstrated that minimizing path curvature directly reduces global truncation errors, enabling high-fidelity generation with drastically fewer function evaluations. This geometric perspective has resolved many inefficiencies inherent in early diffusion models, transforming tortuous stochastic paths into near-linear deterministic flows. However, this pursuit of straightness introduces new challenges, including low-noise pathologies and the potential for trajectory collapse, necessitating sophisticated regularization techniques and hybrid training strategies that balance numerical stability with representational capacity.

The evolution of architectural backbones and control mechanisms further underscores the maturity of this unified landscape. The integration of Transformer-based architectures and State Space Models has overcome historical scalability limitations, allowing flow-based models to compete with diffusion models in high-resolution synthesis while maintaining linear complexity. Simultaneously, controllability has evolved from heuristic guidance to rigorous, geometry-aware frameworks rooted in optimal control theory. Innovations such as adaptive manifold guidance and reward-based fine-tuning via reinforcement learning have enabled precise steering of generative trajectories without violating probability conservation or inducing off-manifold drift. These advancements highlight a paradigm shift where control is achieved not just by altering model weights, but by dynamically manipulating the vector fields and boundary conditions of the generative process.

Despite these significant strides, critical challenges remain in scaling these models to the parameter counts of large language models and extending theoretical guarantees to discrete and non-Euclidean domains. The tension between deterministic efficiency and the stochastic exploration required for robust reinforcement learning alignment continues to drive research into hybrid sampling strategies and variance-aware optimization. Furthermore, ensuring that straightened flows maintain diversity and do not succumb to bias manifolds requires ongoing innovation in orthogonal perturbation and density control methods. Ultimately, the convergence of diffusion and flow paradigms points toward a future of real-time, reasoning-integrated generative systems capable of solving complex scientific and creative tasks with unprecedented speed, fidelity, and controllability.